RAGFailBench revision: tables and figure

Model evaluation (Tables XII, XIII, XV, XVI)

Table 1: Score by condition, mean ± std over three seeded runs. Score is answer accuracy for answer-expected conditions (conflict: handled rate) and correct-abstention rate for abstain-expected ones; closed-book is accuracy with no context (prior knowledge, a probe outside FRS). Δ is the paired drop in percentage points against each item's own parent clean seed; FRS = 1 − mean paired drop.

Condition	nova-pro score (Δ pp)	llama score (Δ pp)	gpt-oss score (Δ pp)
clean	0.98 ± 0.01	0.99 ± 0.01	0.98 ± 0.01
missing_evidence (abstain)	0.92 ± 0.02 (-8.0)	0.93 ± 0.02 (-6.9)	0.93 ± 0.01 (-7.2)
hard_negative (abstain)	0.98 ± 0.01 (-2.3)	0.98 ± 0.01 (-2.1)	0.97 ± 0.01 (-2.6)
conflict	0.70 ± 0.03 (-28.5)	0.72 ± 0.02 (-26.6)	0.68 ± 0.02 (-30.4)
context_noise	0.98 ± 0.01 (-0.1)	0.99 ± 0.00 (+0.1)	0.99 ± 0.01 (+0.3)
chunk_boundary	0.96 ± 0.01 (-1.8)	0.97 ± 0.01 (-1.3)	0.93 ± 0.00 (-5.4)
evidence_position	0.98 ± 0.00 (+0.1)	0.99 ± 0.00 (+0.1)	0.98 ± 0.01 (-0.1)
closed-book (probe)	0.28 ± 0.03	0.47 ± 0.02	0.49 ± 0.03
FRS	0.932 ± 0.003	0.939 ± 0.008	0.924 ± 0.006

Table 2: Abstention / answered-correctly / hallucination rate by condition, mean ± std over three seeded runs. For abstain-expected operators a non-abstention is either correct (prior knowledge or a residual partial leak) or a hallucination (wrong answer).

Condition	nova-pro abst. / ans.-correct / halluc.	llama abst. / ans.-correct / halluc.	gpt-oss abst. / ans.-correct / halluc.
missing_evidence	0.92 ± 0.02 / 0.01 ± 0.01 / 0.07 ± 0.01	0.93 ± 0.02 / 0.01 ± 0.00 / 0.06 ± 0.01	0.93 ± 0.01 / 0.01 ± 0.01 / 0.06 ± 0.01
hard_negative	0.98 ± 0.01 / 0.00 ± 0.00 / 0.02 ± 0.01	0.98 ± 0.01 / 0.00 ± 0.00 / 0.02 ± 0.01	0.97 ± 0.01 / 0.00 ± 0.00 / 0.02 ± 0.01
conflict	0.01 ± 0.00 / — / 0.30 ± 0.03	0.01 ± 0.01 / — / 0.27 ± 0.01	0.01 ± 0.00 / — / 0.31 ± 0.02
context_noise	0.00 ± 0.00 / — / 0.02 ± 0.00	0.00 ± 0.00 / — / 0.01 ± 0.01	0.00 ± 0.00 / — / 0.01 ± 0.00
chunk_boundary	0.02 ± 0.00 / — / 0.02 ± 0.01	0.01 ± 0.01 / — / 0.01 ± 0.01	0.06 ± 0.00 / — / 0.02 ± 0.00
evidence_position	0.00 ± 0.00 / — / 0.02 ± 0.00	0.00 ± 0.00 / — / 0.01 ± 0.00	0.00 ± 0.00 / — / 0.01 ± 0.00

Table 3: Paired drop (pp) by severity: mean ± std over three seeded runs of the three-model mean. For abstain-expected operators the drop is the rate of failing to abstain; missing_evidence severity is closeness of the remaining context to the removed evidence.

Operator	low	medium	high
missing_evidence	0.0 ± 0.0	7.8 ± 2.3	15.7 ± 2.6
hard_negative	1.7 ± 1.3	2.2 ± 0.5	3.1 ± 0.9
conflict	12.2 ± 1.7	28.9 ± 2.0	44.4 ± 2.1
context_noise	-0.1 ± 0.6	-0.1 ± 0.8	-0.1 ± 0.8
chunk_boundary	0.4 ± 0.7	3.0 ± 0.3	5.8 ± 1.2
evidence_position	-0.2 ± 0.6	0.1 ± 0.6	-0.1 ± 0.9

Table 4: Conflict outcomes, mean ± std (%) over three seeded runs. Handled = gold, both values, or an acknowledged conflict. Aware = prompt that permits flagging a conflict (probe, excluded from FRS). Prior known/unknown splits by closed-book correctness.

Model	Prompt	Gold %	Alternate %	Both %	Acknowledged %	Handled %	Judge-only %	Handled, prior known %	Handled, prior unknown %
nova-pro	default	59.3 ± 2.4	25.8 ± 4.2	10.4 ± 1.3	10.6 ± 1.5	69.7 ± 3.4	69.2 ± 3.2	86.5 ± 4.0	63.2 ± 3.0
nova-pro	aware	30.5 ± 1.7	9.3 ± 1.6	55.8 ± 2.1	58.5 ± 2.0	88.6 ± 0.8	37.1 ± 1.2	95.0 ± 2.8	86.1 ± 0.1
llama	default	36.8 ± 1.9	18.2 ± 1.6	34.9 ± 1.2	35.1 ± 1.4	72.0 ± 2.3	62.1 ± 2.8	78.0 ± 3.7	66.6 ± 4.3
llama	aware	13.7 ± 0.4	2.3 ± 0.3	77.1 ± 0.8	83.9 ± 0.2	95.5 ± 0.3	31.3 ± 2.9	97.5 ± 0.2	93.8 ± 0.6
gpt-oss	default	50.4 ± 1.9	25.2 ± 2.2	17.5 ± 2.5	17.6 ± 2.7	67.9 ± 2.1	64.2 ± 2.6	78.2 ± 3.0	58.1 ± 3.9
gpt-oss	aware	13.6 ± 2.5	2.6 ± 0.4	74.5 ± 5.7	84.9 ± 3.2	95.6 ± 0.8	23.2 ± 3.4	95.5 ± 1.1	95.7 ± 1.1

Table 5: Paired drop (pp) and FRS with 95% CIs, pooled over three seeded runs; cluster bootstrap over 561 parent clean seeds, 2000 resamples.

Condition	nova-pro	llama	gpt-oss
missing_evidence	8.0 [6.6, 9.6]	6.8 [5.4, 8.3]	7.2 [5.8, 8.7]
hard_negative	2.3 [1.3, 3.4]	2.1 [1.3, 3.1]	2.6 [1.5, 3.8]
conflict	28.5 [25.6, 31.4]	26.6 [24.1, 29.3]	30.4 [27.6, 33.3]
context_noise	0.1 [-0.9, 1.0]	-0.1 [-0.9, 0.7]	-0.3 [-1.0, 0.4]
chunk_boundary	1.8 [0.7, 2.8]	1.2 [0.2, 2.4]	5.4 [3.8, 7.0]
evidence_position	-0.1 [-1.1, 0.8]	-0.1 [-0.7, 0.4]	0.1 [-0.6, 0.8]
FRS	0.932 [0.925, 0.940]	0.939 [0.933, 0.945]	0.924 [0.917, 0.931]

Table 6: FRS difference between models with paired 95% CIs (same resampled seeds), pooled over three seeded runs; cluster bootstrap over 561 parent clean seeds, 2000 resamples.

Models	ΔFRS [95% CI]	CI excludes 0
nova-pro – llama	-0.006 [-0.015, +0.002]	no
nova-pro – gpt-oss	+0.008 [-0.000, +0.016]	no
llama – gpt-oss	+0.015 [+0.006, +0.022]	yes

Confidence intervals, McNemar tests, correctness signals (Tables XVII, XIX)

Table 7: Exact McNemar tests between models on the same items (expected-behavior score; pooled over three seeded runs). Discordant = items only the first / only the second model gets right; p (Holm) corrects over all 21 tests.

Condition	Models	n	Δ pp	Discordant	p	p (Holm)
clean	nova-pro vs llama	561	-0.4	4 / 6	0.75	1
clean	nova-pro vs gpt-oss	561	+0.0	5 / 5	1	1
clean	llama vs gpt-oss	561	+0.4	4 / 2	0.69	1
missing_evidence	nova-pro vs llama	1552	-1.2	28 / 46	0.047	0.85
missing_evidence	nova-pro vs gpt-oss	1552	-0.8	32 / 44	0.21	1
missing_evidence	llama vs gpt-oss	1552	+0.4	37 / 31	0.54	1
hard_negative	nova-pro vs llama	1605	-0.2	13 / 16	0.71	1
hard_negative	nova-pro vs gpt-oss	1605	+0.3	16 / 11	0.44	1
hard_negative	llama vs gpt-oss	1605	+0.5	21 / 13	0.23	1
conflict	nova-pro vs llama	1683	-2.3	188 / 226	0.069	1
conflict	nova-pro vs gpt-oss	1683	+1.8	235 / 204	0.15	1
conflict	llama vs gpt-oss	1683	+4.1	241 / 172	0.0008	0.015
context_noise	nova-pro vs llama	1683	-0.5	17 / 25	0.28	1
context_noise	nova-pro vs gpt-oss	1683	-0.4	15 / 21	0.41	1
context_noise	llama vs gpt-oss	1683	+0.1	17 / 15	0.86	1
chunk_boundary	nova-pro vs llama	1368	-0.8	15 / 26	0.12	1
chunk_boundary	nova-pro vs gpt-oss	1368	+3.8	70 / 18	2.1e-08	4.2e-07
chunk_boundary	llama vs gpt-oss	1368	+4.6	73 / 10	5.8e-13	1.2e-11
evidence_position	nova-pro vs llama	1683	-0.4	15 / 21	0.41	1
evidence_position	nova-pro vs gpt-oss	1683	+0.2	22 / 18	0.64	1
evidence_position	llama vs gpt-oss	1683	+0.6	18 / 8	0.076	1

Table 8: Share of answer-expected items each correctness signal accepts, pooled over three seeded runs. The reported answer accuracy is the union (exact match, token-F1 ≥ 0.6, or judge). Conflict here is answer correctness before outcome scoring.

Condition	Model	Exact match	Token-F1 ≥ 0.6	Judge	Union (reported)
clean	nova-pro	0.731	0.879	0.971	0.982
clean	llama	0.745	0.884	0.975	0.986
clean	gpt-oss	0.688	0.843	0.973	0.982
conflict	nova-pro	0.434	0.603	0.595	0.692
conflict	llama	0.219	0.497	0.442	0.621
conflict	gpt-oss	0.295	0.516	0.504	0.642
context_noise	nova-pro	0.730	0.875	0.970	0.982
context_noise	llama	0.747	0.887	0.976	0.986
context_noise	gpt-oss	0.687	0.847	0.976	0.985
chunk_boundary	nova-pro	0.751	0.874	0.962	0.965
chunk_boundary	llama	0.773	0.902	0.969	0.973
chunk_boundary	gpt-oss	0.689	0.819	0.925	0.927
evidence_position	nova-pro	0.736	0.877	0.973	0.983
evidence_position	llama	0.747	0.891	0.977	0.987
evidence_position	gpt-oss	0.693	0.853	0.973	0.981

Table 9: Human audit summary (replaces the HAR and failure-validity rows of Table V). Blind stratified sample from the three seeded runs; 95% Wilson CIs. Annotator 1 is primary.

Metric	Annotator 1	Annotator 2
Human acceptance (HAR), clean seeds	84.0% [71.5, 91.7] (42/50)	86.0% [73.8, 93.0] (43/50)
Failure injection valid	96.3% [87.5, 99.0] (52/54)	98.1% [90.2, 99.7] (53/54)
Failure label (answer available) correct	100.0% [93.4, 100.0] (54/54)	100.0% [93.4, 100.0] (54/54)

Table 10: Human audit per seeded run (replaces Tables VI and VIII human columns): clean seeds kept and failure cases judged valid, per annotator. Per-run samples are small (16–17 seeds, 18 failure cases); pooled values are in the summary table.

Run	HAR A1	HAR A2	Failure validity A1	Failure validity A2
pilot_stability_s42	16/16	16/16	17/18	17/18
pilot_stability_s123	13/17	13/17	17/18	18/18
pilot_stability_s2026	13/17	14/17	18/18	18/18
All runs	42/50	43/50	52/54	53/54

Human audit (Tables V, VI, VIII, IX, XI human rows)

Table 11: Primary reason for the 8 of 50 audited clean seeds annotator 1 did not keep (replaces Table IX). One primary reason per row from the structured flags, in the order listed; the free-text note gives the detail.

Reason	Count	Percent	Structured basis	Rows
Time-sensitive phrasing	2	25.0%	time_sensitive_ok = no	C022 (fix), C027 (fix)
Answer not unique	1	12.5%	answer_unique = no	C002 (fix)
Unclear question (reference, premise, wording)	4	50.0%	question_clear = no	C010 (fix), C039 (fix), C044 (fix), C047 (fix)
Other (trivial; see note)	1	12.5%	no flag set	C040 (reject)

Table 12: Failure-injection verification: automatic check on every case versus the human audit (replaces Table XI). Per-operator samples are 9 cases (3 per severity).

Check	Annotator 1	Annotator 2
Automatic structural verification (leakage / label check), all cases	94.9% ± 0.3% pass	
Human: injection valid, all operators	96.3% [87.5, 99.0] (52/54)	98.1% [90.2, 99.7] (53/54)
missing_evidence	9/9	9/9
hard_negative	9/9	9/9
conflict	7/9	8/9
context_noise	9/9	9/9
chunk_boundary	9/9	9/9
evidence_position	9/9	9/9

Table 13: Inter-annotator agreement on the same audit rows. The second annotator had seen the first annotator's labels in an earlier pass, so agreement may be overstated. Kappa is undefined when both annotators used a single identical label.

Field	n	Agreement	Cohen's kappa
decision	50	98.0%	0.92
question_clear	50	98.0%	0.85
answer_in_evidence	50	100.0%	undefined (one label only)
answer_unique	50	98.0%	0.66
time_sensitive_ok	50	100.0%	1.00
needs_title	50	100.0%	1.00
human_injection_valid	54	98.1%	0.66
human_label_correct	54	100.0%	undefined (one label only)
severity_ok	54	74.1%	0.14

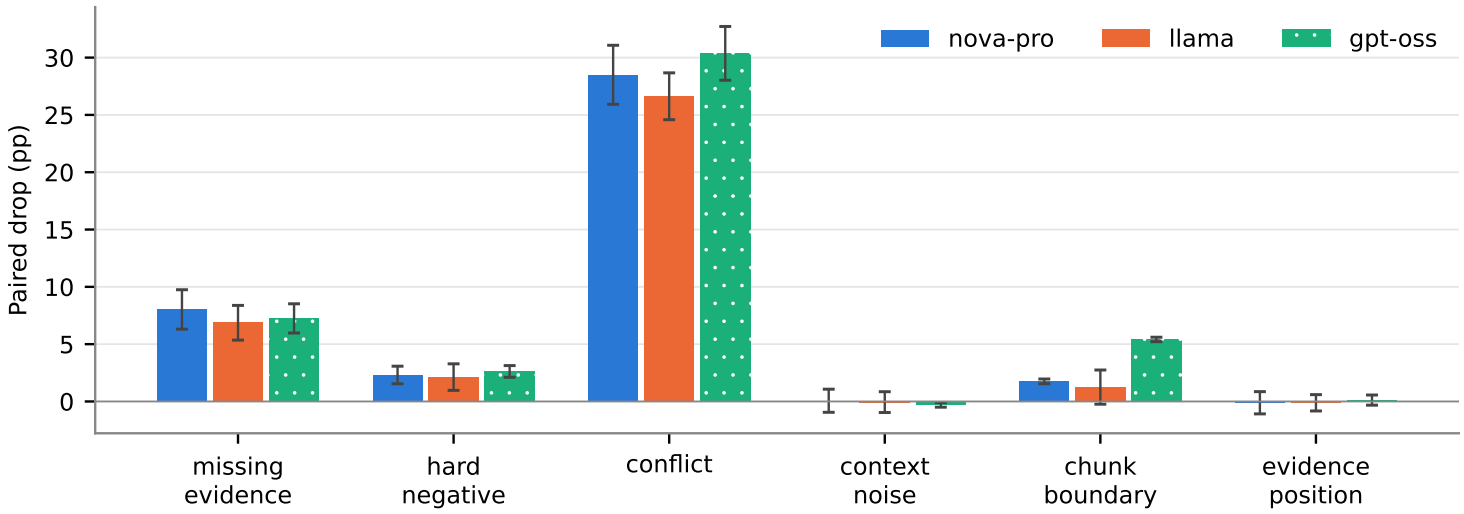


Figure 1: Paired performance drop by failure operator (Fig. 2).